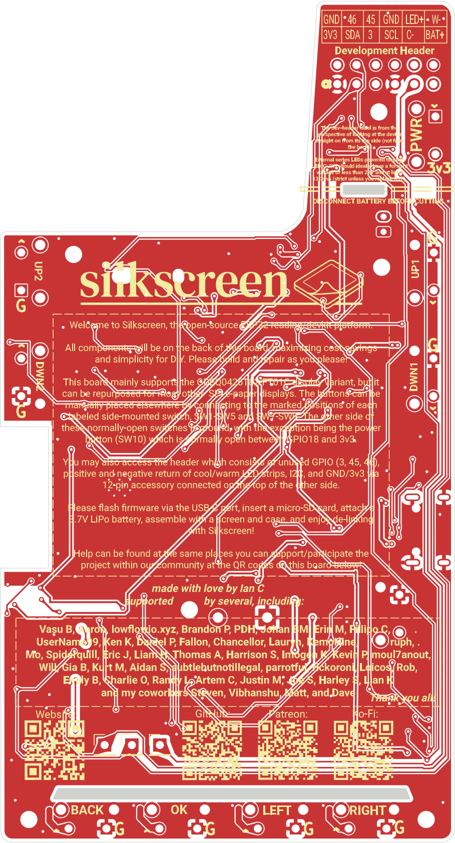


The dev-header label is from the perspective of looking at the device straight on from its top side (not from the back).

External series LEDs powered through LED/C-/W- should ideally have a forward voltage of less than 22V and at least 13.3mA (strict unless you replace R37).

Welcome to Silkscreen, the open-source ESP32 reading/devkit platform.

All components will be on the back of this board, maximizing cost-savings and simplicity for DIY. Please build and repair as you please!

This board mainly supports the GDEQ0426T82-FT01C display variant, but it can be repurposed for most other SPI e-paper displays. The buttons can be manually placed elsewhere by connecting to the marked positions of each labeled side-mounted switch, SW1-SW5 and SW7-SW9. The other side of these normally-open switches is ground, with the exception being the power button (SW10) which is normally open between GPIO18 and 3v3.

You may also access the header which consists of unused GPIO (3, 45, 46), positive and negative return of cool/warm LED strips, I2C, and GND/3v3 via 12-pin accessory connected on the top of the other side.

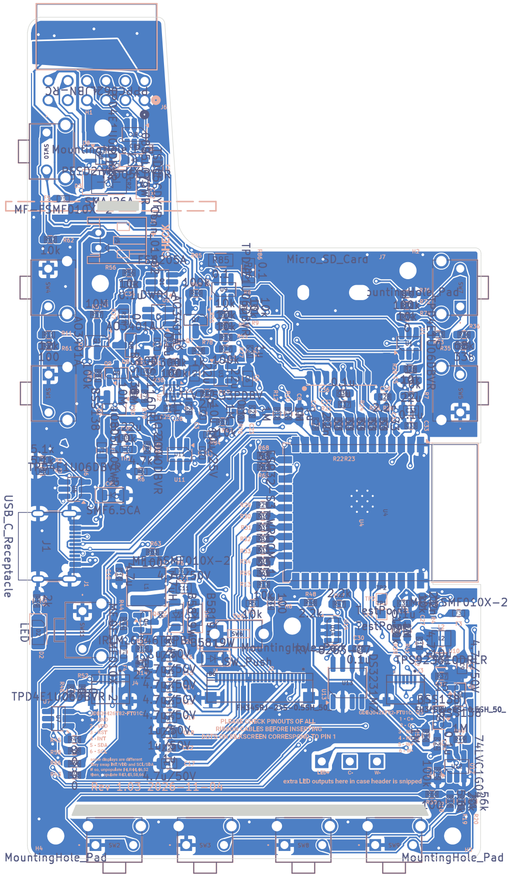
Please flash firmware via the USB-C port, insert a micro-SD card, attach a 3.7V LiPo battery, assemble with a screen and case, and enjoy de-linking with Silkscreen!

Help can be found at the same places you can support/participate the project within our community at the QR codes on this board below!

made with love by Ian C
supported by several, including:

Vasu B, Aaron, lowflowio.xyz, Brandon P, PDH, Johan BM, Erin M, Fillipo C, UserName99, Ken K, Daniel P, Fallon, Chancellor, Lauryn, KemoNine, ruph, Mo, Spiderquill, Eric J, Liam H, Thomas A, Harrison S, Imogen K, Kevin P, moul7anout, Will, Gia B, Kurt M, Aidan S, subtlebutnotillegal, parrotful, rickoroni, Loicos, Rob, Emily B, Charlie O, Randy L, Artem C, Justin M, Joe S, Harley S, Lian K and my coworkers Steven, Vibhanshu, Matt, and Dave

Thank you all!


Micro_SD_Card
USB_C_Receptacle
LED
MountingHole_Pad
MountingHole_Pad
MF-FSMF010X
SMAJ26A
SMF6.5CA
MF-FSMF010X-2
IRLML6346TRPBF
B5819W
SW_Push
TPS92360DRLR
BSS138
74LVC1G04
FH34SRJ-24S-0.5SH_50_
GDEQ0426T82-FT01C:
1 - GND
2 - VDD
3 - RST
4 - INT
5 - SDA
6 - SCL
some displays are different
they swap INT/VDD and SCL/SDA
If so, unpopulate F4,R44,46,52
then, populate R43,45,58,66
GDEQ0426T82-FT01C:
1 - C+
2 - C-
3 - NC
4 - NC
5 - W+
6 - W-
PLEASE CHECK PINOUTS OF ALL
RIBBON CABLES BEFORE INSERTING
DOTS ON SILKSCREEN CORRESPOND TO PIN 1
LED+
C-
W-
extra LED outputs here in case header is snipped
Rev 1.03 2026-11-04
SW2
SW3
SW8
SW9
SW10
4.7u/50V
1u/50V
0.1u
10k
100k
1M
2.2k
56k
100
DS3231MZ
U4
U11
U13
J1
J2
J4
J7